

Assignment – 5

Topics Covered: Definition, difference of means: independent samples (t-test), paired samples t-test, F-test, definition, independence of attributes, introduction, assumptions, classification of ANOVA, one-way ANOVA.

A1. An F-test is typically used to compare the of two or more groups to determine if they differ significantly.

A2. What does ANOVA (Analysis of Variance) help us test?

A3. In a paired samples t-test, each observation in one group iswith an observation in the other group.

A4. What is the null hypothesis in a paired samples t-test?

A5. The alternative hypothesis in a paired samples t-test states that the mean difference between paired observations is

A6. To test independence of attributes, the observed data are arranged in a table.

A7. How is the F-ratio calculated in an F-test?

B1. Two types of batteries, X and Y are tested for their length of life and the following results are obtained:

Battery	Sample size	Mean (hours)	Variance (hours)
A	10	1000	100
B	12	2000	121

Is there a significant difference between the two means? (Tabulated value: 1.325)

B2. A travel agency's marketing brochure indicates that the standard deviations of hotel rates for two cities are the same. A random sample of 13 rooms rates in a city has a standard deviation of \$ 27.50 and a random sample of 16 hotel room rates in the other city has a standard deviation of \$ 29.75. Can you reject the agency's claim at $\alpha = 0.01$? (Tabulated value: 4.72)

B3. In a certain sample of 2,000 families, 1,400 people are consumers of tea. Out of 1,800 Hindu families, 1,236 families are consumers of tea. State whether there is any significant difference between consumption of tea among Hindu and non-Hindu families. (Tabulate value: 3.841).

B4. A certain oil corporation a seismic test on land assess the possibilities of extraction of oil from the land. From the past 200 records, the following test results and oil yields were obtained

Test Results	Oil yield Frequencies		
	High	Medium	Low
Good	44	22	13
Fair	15	19	26
Bad	6	8	47

Can we say, 1% level of significance, that oil yields and test results are associated?

Tabulated value= 13.28

C1. In a test given to two groups of students, the marks obtained are as follows:

First Group	18	20	36	50	49	36	34	49	41
Second Group	29	28	26	35	30	44	46		

Test if the average marks secured by two groups of students differ significantly. (Tabulated value: 1.345)

C2. Two random samples were drawn from two normal populations and their values are as follows:

A	66	67	75	76	82	84	88	90	92		
B	64	66	74	78	82	85	87	92	93	95	97

Test whether the two populations have the same variance at 10% level of significance. (Tabulated value: 3.07)

C3. 1,000 families were selected at random in a city to test the belief that the high-income group families usually send their children to public schools and the low income families often send their children to government schools. The following results were obtained:

Income	Public School	Government School	Total
Low	370	430	800
High	130	70	200
Total	500	500	1000

Test whether income and type of schooling are independent. (Tabulated value: 3.841)

C4. Three different techniques, namely medication, exercises and special diet are randomly assigned to (individuals diagnosed with high blood pressure) lower blood pressure. After four weeks the reduction in each person's blood pressure is recorded. Test at 5% level, whether there is significant difference in mean reduction of blood pressure among the three techniques. (Tabulated value: 19.41)

Medication	10	12	9	15	13
Exercise	6	8	3	0	2
Diet	5	9	12	8	4

D1. For a random sample of 10 persons fed on diet A the increased weight in pounds in a certain period were 10, 6, 16, 17, 13, 12, 8, 14, 15, and 9. For another random sample of 12 persons fed on diet B, the increase in the same period were 7, 13, 22, 15, 12, 14, 18, 8, 21, 23, 10, and 17. Test whether the diets A and B differ significantly as regards their effect on increase in weight. (Tabulate value: 1.325)

D2. Consider the following measurements of the heat producing capacity of the coal produced by two mines (in millions of calories per ton):

Mine 1	8260	8130	8350	8070	8340	
Mine 2	7950	7890	7900	8140	7920	7840

Can it be concluded that the two population variances are equal? (Tabulate value: 5.19)

D3. Three composition instructors recorded the number of spelling errors which their students made on a research paper. At 1% level of significance test whether there is significant difference in the average number of errors in the three classes of students. (Tabulated value:19.43)

Instructor 1	2	3	5	0	8		
Instructor 2	4	6	8	4	9	0	2
Instructor 3	5	2	3	2	3	3	

D4. The sales data of an item in six shops before and after a special promotional campaign are as under:

Shops	A	B	C	D	E	F
Before Campaign	53	28	31	48	50	42
After Campaign	58	29	30	55	56	45

Can the campaign be judged to be a success? Test at 5% level of significance.
(Tabulated Value=2.02)